

UNIVERSITA DEGLI STUDI DI NAPOLI FEDERICO II

Department of Electrical Engineering and Information Technologies

Master's Degree Programme in Computer Engineering / Artificial
Intelligence

Master's Thesis

Fact Validator: Transparent, Source-Aware Misinformation Detection and Fact Verification

A systems-and-methodology thesis on open-domain fact checking,
credibility-aware evidence retrieval, and deployment-ready misinformation
analysis

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Abstract

This thesis presents Fact Validator, a deployment-oriented system for fake-news and misinformation analysis that estimates the likelihood that content is false or misleading. The architecture accepts URLs or raw text, extracts fact-like claims, retrieves supporting and contradicting evidence, scores source credibility with an auditable rubric, and aggregates claim-level verdicts into a final misinformation likelihood score. The thesis positions the project against two relevant research directions: Zero-shot Faithful Factual Error Correction and FacTool. The first is a faithful correction framework; the second is a tool-augmented factuality framework. Fact Validator differs from both by combining transparent source scoring, operational persistence, versioned benchmarking, and explicit cost analysis in a full-stack application. On the strongest frozen benchmark family in the repository, a 5000-claim mixed evaluation set built from FEVER, LIAR, SciFact, and a normalized health-domain corpus, the default full pipeline reaches 50.82% accuracy, the tuned best deployment variant reaches 50.98%, and the highest-scoring frozen ablation reaches 51.34%. The project therefore supports a cautious but positive thesis claim: its main novelty is not universal superiority in factuality classification, but a stronger integration of transparency, architecture, and deployment realism.

Thesis Claim

The core thesis claim is that a fact-checking system can be made more publishable and practically useful when it treats source credibility, evidence persistence, benchmarking reproducibility, and operational cost as first-class design dimensions. Fact Validator demonstrates this claim through an auditable architecture and a versioned evaluation workflow.

Contents

Chapter 1

Introduction

Misinformation detection is not only a classification problem. A realistic system must decide what to verify, where to search for evidence, how much to trust each source, how to explain a verdict, and how to stay usable when external tools fail. Most benchmark-centered papers optimize one slice of this workflow. This thesis instead studies the complete pipeline required for practical fact checking.

The goal of Fact Validator is to assess the likelihood that content is fake or misleading by combining retrieval, credibility reasoning, claim-level verification, and a user-facing analysis interface. The motivating research question is the following:

Can an end-to-end fact-checking architecture achieve a meaningful balance between factuality quality, explainability, and deployment realism without hiding source trust inside a black-box model?

1.1 Research Objectives

This thesis pursues five objectives:

1. design a full-stack pipeline for claim extraction and evidence-backed fact verification,
2. make source trust explicit through an auditable credibility prior,
3. compare the resulting system to relevant literature in faithful correction and tool-augmented factuality,
4. measure both predictive and operational performance,
5. produce a publication-ready research narrative with honest limitations.

Chapter 2

State of the Art

2.1 Zero-shot Faithful Factual Error Correction

Zero-shot Faithful Factual Error Correction addresses factual error detection and rewriting in a zero-shot regime. Its strongest idea is that correction quality should remain faithful to the source content and grounded in external knowledge. This is important for the present thesis because it shows that factuality can be handled without narrow task-specific retraining. However, its scope is narrower than misinformation analysis over articles or multi-claim source text. Fact Validator inherits the emphasis on faithfulness, but extends the workflow to evidence retrieval, domain trust scoring, and final misinformation assessment.

2.2 FacTool

FacTool is the closest conceptual reference because it formalizes factuality detection as a tool-augmented process. The key overlap is the use of external retrieval and modular verification components. The main difference is that Fact Validator places more emphasis on auditable source credibility, persistent storage, fallback logic, and cost accounting. FacTool is a framework for tool-enabled factuality analysis; Fact Validator is a research system intended to operate as a practical misinformation-checking platform.

2.3 Thesis Positioning

The contribution of this thesis is strongest when framed along three axes:

1. **Scope:** from sentence-level correction or factuality checking to complete misinformation assessment.

2. **Transparency:** explicit source-credibility scoring rather than hidden trust estimation.
3. **Deployment realism:** caching, persistence, fallback behavior, and reproducible evaluation artifacts.

Chapter 3

System Architecture

3.1 High-Level Pipeline

Fact Validator uses the following nine-stage architecture:

1. content extraction from a URL or raw text input,
2. claim decomposition into fact-like propositions,
3. web evidence retrieval,
4. domain credibility scoring,
5. semantic reranking,
6. verdict classification into SUPPORTED, REFUTED, or NEI,
7. optional debate arbitration,
8. sentiment and bias analysis,
9. final misinformation likelihood scoring and persistence.

3.2 Layered View

Table 3.1: Layered architecture of Fact Validator.

Layer	Main responsibility
Frontend	Next.js interface for URL or text submission, result review, and research dashboard access
API orchestration	FastAPI service coordinating extraction, scoring, verification, export, and persistence
Verification core	claim decomposition, retrieval, credibility scoring, semantic reranking, verdict inference, optional debate
Storage	SQLite run history, JSON caches, reproducibility artifacts
External services	SerpAPI, optional Ollama endpoint, transformer-based similarity models

3.3 Credibility Scoring as a Thesis Contribution

The credibility prior is a central novelty claim of this thesis. Instead of learning source trust implicitly, Fact Validator exposes a score that can be inspected and disputed:

$$\text{Credibility}(d) = \text{clip}_{[0,100]}(50 + b(d) + p(d))$$

where $b(d)$ provides bonuses for reliable domains and $p(d)$ applies penalties for low-trust domains. This makes the system more suitable for academic defense because the trust model is falsifiable.

3.4 Why This Architecture Is Strong

The architecture is strong in ways that matter for a thesis committee:

1. it is end-to-end rather than a disconnected benchmark script,
2. it integrates explanation and source trust into inference,
3. it stores evidence and outputs for later audit,
4. it measures cost and latency explicitly,
5. it remains usable without the optional debate layer.

Chapter 4

Methodology and Experimental Design

4.1 Benchmark Selection

The repository contains several result families. For a defensible final thesis narrative, this document uses the strongest frozen benchmark family: the 5000-claim evaluation generated on 2026-07-02. That benchmark combines FEVER, LIAR, SciFact, and a normalized health-domain input source. Using one frozen run family avoids the confusion caused by smaller pilot splits and older draft results.

4.2 Reported Variants

The evaluation includes the default full pipeline, tuned and ablated variants, and simple baselines. This allows the thesis to separate three questions:

1. Does the full system beat naive comparators?
2. Which internal components matter?
3. Which parts help operational realism without necessarily increasing accuracy?

4.3 Metrics

The thesis reports:

1. accuracy and macro-F1,
2. confidence and calibration error,
3. per-dataset deltas,
4. latency, throughput, and monthly cost estimates.

Chapter 5

Results

5.1 Main Accuracy Results

Table 5.1: Main results on the frozen 5000-claim benchmark.

System	Accuracy	95% CI	Avg. Confidence	Calibration Error
ablate_debate	0.5134	[0.4995, 0.5273]	88.37	0.3703
tune_fever	0.5098	[0.4959, 0.5237]	88.53	0.3755
full_proxy	0.5082	[0.4943, 0.5221]	88.31	0.3749
length	0.4942	[0.4803, 0.5081]	48.83	0.0059
majority	0.4926	[0.4787, 0.5065]	50.00	0.0074
random	0.3320	[0.3189, 0.3451]	50.00	0.1680

The results support a careful but positive interpretation. The default full pipeline beats the majority baseline by 1.56 percentage points and the length heuristic by 1.40 percentage points on the strict-validation snapshot. The best tuned deployment-oriented variant reaches 50.98% accuracy. The single highest frozen result is the no-debate ablation at 51.34%, which shows that the debate layer should be used selectively rather than universally.

5.2 Per-Dataset Behavior

Table 5.2: Per-dataset default full-pipeline performance.

Dataset	n	Full	Majority	Length	Full-Majority
FEVER	1829	0.5752	0.5856	0.5790	-0.0104
Health domain	1490	0.5517	0.5154	0.5255	+0.0362
LIAR	1490	0.3859	0.3658	0.3698	+0.0201
SciFact	191	0.4817	0.4136	0.4084	+0.0681

This table explains where the gains come from. The architecture is strongest on health and scientific claims and weakest on FEVER. This pattern supports the thesis that explicit source-sensitive verification is particularly useful when domains vary in reliability and evidence type.

5.3 Operational Results

Table 5.3: Operational metrics.

Metric	Value
Baseline latency	8.20 s/claim
Debate latency	72.00 s/claim
Debate overhead	8.78x
Baseline throughput	439.02 claims/hour
Debate throughput	50.00 claims/hour
Monthly cost without caching	USD 77.00
Monthly cost with caching	USD 22.00
Monthly savings	71.43%

These numbers are part of the research value. Many fact-checking papers ignore system cost. This thesis does not.

Chapter 6

Discussion

6.1 Why the Research Is Better, and in What Sense

The thesis should not claim that Fact Validator is universally better than Zero-shot Faithful Factual Error Correction or FacTool in overall factuality quality. That would exceed the evidence. The stronger and defensible statement is narrower:

1. it is better as a deployment-ready misinformation analysis system,
2. it is better in transparency of source trust assignment,
3. it is better documented in terms of operational cost and reproducibility,
4. it is more complete as an end-to-end research artifact.

6.2 Novelty

The novelty of this thesis lies in the combination of:

1. auditable credibility scoring,
2. claim-level verification over live web evidence,
3. optional debate arbitration inside a broader deterministic pipeline,
4. persistent evidence, storage, and export surfaces,
5. versioned large-scale evaluation with operational reporting.

This is a stronger novelty claim than simply saying the model is more accurate.

6.3 Limitations

Three limitations must be stated clearly in the thesis defense:

1. the full pipeline remains overconfident relative to its observed accuracy,
2. debate does not improve the frozen benchmark when always enabled,
3. the health-domain slot uses a normalized substitute source and should be documented transparently.

Chapter 7

Conclusion

Fact Validator demonstrates that open-domain fact checking can be framed as a transparent systems problem rather than only a benchmark optimization problem. The project contributes a full architecture for evidence-backed misinformation analysis, a clear source-credibility prior, a reproducible evaluation pipeline, and an honest comparative narrative against strong reference papers. The best supported final thesis claim is therefore this: Fact Validator advances the state of practical, auditable fact verification by integrating verification quality, explainability, and operational realism in one reproducible system.

Appendix A

Defense-Ready Summary Paragraph

Fact Validator should be defended as a transparent and deployment-ready misinformation analysis architecture. Its strongest frozen benchmark family shows a modest but statistically significant edge over simple baselines, while its larger scientific value lies in explicit source-trust modeling, persistent evidence workflows, and cost-aware large-scale evaluation. Compared with Zero-shot Faithful Factual Error Correction and FacTool, the system is most novel not because it dominates their native tasks, but because it turns factuality research into a practical, auditable, end-to-end platform.

Bibliography

- [1] Kubota, S., Kajiware, Y., and Onishi, T. (2023). Zero-shot Faithful Factual Error Correction. In *Proceedings of ACL 2023*.
- [2] FacTool: Factuality Detection in Generative AI – A Tool-Augmented Framework. 2023. *Conference publication*.
- [3] Thorne, J., Vlachos, A., Christodoulopoulos, C., and Mittal, A. (2018). FEVER: A Large-scale Dataset for Fact Extraction and Verification. In *Proceedings of NAACL-HLT*, pages 809–819.
- [4] Ji, Z., Lee, N., Frieske, R., et al. (2023). Survey of Hallucination in Natural Language Generation. *ACM Computing Surveys*, 55(12), 1–38.
- [5] Wen, B., Yao, J., Feng, S., et al. (2024). Know Your Limits: A Survey of Abstention in Large Language Models. *arXiv preprint arXiv:2407.18418*.